

Garrett G. Wen

Ph.D. Candidate
Department of Statistics and Data Science
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Google Scholar

Education

- **Yale University** **New Haven, CT, USA**
Ph.D. candidate, Department of Statistics and Data Science. 09/2022 – Present
Ph.D. Advisors: Prof. Theodor Misiakiewicz and Prof. Zhou Fan.
Expected graduation / degree conferral: May 24, 2027.
M.A. in Statistics (en route), conferred 12/2025.
- **Peking University** **Beijing, China**
B.Sc. in Statistics, School of Mathematical Sciences. 09/2018 – 06/2022

Academic & Research Experience

- **Flatiron Institute, Center for Computational Mathematics (CCM)** **New York, NY**
Summer Research Associate. 06/01/2026 – 08/14/2026
Advisors: Dr. Alberto Bietti, Dr. Nikhil Ghosh, and Dr. Denny Wu.
- **The Wharton School, University of Pennsylvania** **Philadelphia, PA**
Visiting Researcher, Department of Statistics and Data Science. 05/2023 – 07/2023
07/2024 – 08/2024; 05/2025 – 07/2025
Remote Research Intern, Department of Statistics and Data Science. 03/2021 – 02/2022
Research on deep learning theory and peer-review mechanisms.
Advisor: Prof. Weijie Su.
- **MEGVII, Beijing, China** 02/2022 – 08/2022
Research Intern, Automated Theorem Proving with Deep Learning.
Advisor: Dr. Shuchang Zhou.
- **Peking University, Beijing, China** 08/2019 – 02/2022
Undergraduate Researcher, School of Mathematical Sciences.
Advisor: Prof. Zhihua Zhang.

Summer Schools

- **Statistical Physics & Machine Learning: Moving Forward**, Cargèse, France 08/2025
- **Princeton Machine Learning Theory Summer School**, Princeton, NJ 06/2023

Research Interests

- Deep learning theory, non-convex optimization, and high-dimensional learning dynamics
- High-dimensional statistics, probability theory, and random matrix theory
- Generative AI, including language-model watermarking and synthetic data
- Mechanism design for scientific evaluation and peer review

Awards & Honors

- Yale Graduate Student Assembly Conference Travel Fellowship 10/2025
- Award for Academic Excellence, Peking University 2019, 2020, 2021, 2022
- **First Prize**, 11th Chinese Mathematics Competition for College Students, National Final (Senior Division) 04/2021
- Guangzhou Pharmaceutical Wanglaoji Scholarship, Peking University 2021
- CreditEase Internet Finance Scholarship, Peking University 2020

Teaching Experience

- **Teaching Fellow (TF) at Yale University** Fall 2024 – Spring 2026
 - S&DS 312/S&DS 612: Linear Models Fall 2024/Fall 2025
Instructor: Prof. Zongming Ma.
 - S&DS 361/AMTH 361/S&DS 661: Data Analysis Spring 2025
Instructor: Prof. Brian Macdonald.
 - S&DS 6590: Mathematics of Deep Learning Spring 2026
Instructor: Prof. Theodor Misiakiewicz. Delivered one course lecture on benign overfitting in February 2026.

Publications

* denotes equal contribution; in (J1), the two co-first authors are listed with Gang Wen first.

Conference and Workshop Papers:

- (C2) “Fast Learning Rate Transfer for Gradient Descent in Sketched Linear Regression.” Garrett Wen, Alberto Bietti, Nikhil Ghosh, Theodor Misiakiewicz, Denny Wu. *ICML Workshop on High-Dimensional Learning Dynamics (HiLD)*, **spotlight**, 2026.
- (C1) “PeriodicSketch: Finding periodic items in data streams.” Zhuochen Fan, Yinda Zhang, Tong Yang, Mingyi Yan, Gang Wen, Yuhan Wu, Hongze Li, Bin Cui. *2022 IEEE 38th International Conference on Data Engineering (ICDE)*, pp. 96–109, 2022.

Journal Papers:

- (J4) “Recommending Best Paper Awards via the Isotonic Mechanism.”
Garrett G. Wen, Buxin Su, Natalie Collina, Zhun Deng, Weijie Su.
Accepted for publication in *Journal of Machine Learning Research*, 2026.
- (J3) “Optimal Estimation of Watermark Proportions in Hybrid AI-Human Texts.”
Xiang Li, Garrett Wen, Weiqing He, Jiayuan Wu, Qi Long, Weijie J. Su.
Minor revision at *Journal of the American Statistical Association*, 2026.
- (J2) “Self-rankings as a predictor of scientific impact beyond peer review.”
Buxin Su, Natalie Collina, Garrett Wen, Didong Li, Kyunghyun Cho, Jianqing Fan, Bingxin Zhao, Weijie Su.
Nature Computational Science, 2026. DOI: [10.1038/s43588-026-01039-0](https://doi.org/10.1038/s43588-026-01039-0).
- (J1) “On the Evolutionary of Bloom Filter False Positives—An Information Theoretical Approach to Optimizing Bloom Filter Parameters.”
Gang Wen*, Zhuochen Fan*, Zhipeng Huang, Yang Zhou, Qiaobin Fu, Tong Yang, Alex X. Liu, Bin Cui.
IEEE Transactions on Knowledge and Data Engineering, 35(7), pp. 7316–7327, 2023.

Preprints

** denotes alphabetical order.

- (P4) “The sharp SAT/UNSAT phase transition in random ellipsoid fitting.”
Theodor Misiakiewicz**, Garrett G. Wen**.
Submitted to *The Annals of Probability*; arXiv:2608.10184, 2026.
- (P3) “Dynamical mean-field limit and replica-symmetric free energy for the orthogonally-invariant SK model.”
Zhou Fan**, Theodor Misiakiewicz**, Leda Wang**, Garrett G. Wen**.
Submitted to *The Annals of Probability*; arXiv:2607.10102, 2026.
- (P2) “When does Gaussian equivalence fail and how to fix it: Non-universal behavior of random features with quadratic scaling.”
Garrett G. Wen, Hong Hu, Yue M. Lu, Zhou Fan, Theodor Misiakiewicz.
Preprint arXiv:2512.03325, 2025.
- (P1) “Optimal Detection for Language Watermarks with Pseudorandom Collision.”
T. Tony Cai**, Xiang Li**, Qi Long**, Weijie J. Su**, Garrett G. Wen**.
Preprint arXiv:2510.22007, 2025.

Selected Talks and Presentations

- (T3) *International Conference on Statistics and Data Science (ICSIDS)*, IMS, Seville 12/2025
- (T2) “High-Dimensional Limit Theorems for SGD: Effective Dynamics and Critical Scaling,”
Machine Learning Theory Reading Group, Yale University 09/2024
- (T1) “The Isotonic Mechanism for Recommending Best Paper Awards,” poster presentation,
Workshop on Incentives in Academia (WINA-24), New Haven 07/2024

Service

- **Reviewer:** Neural Information Processing Systems (NeurIPS), International Conference on Learning Representations (ICLR), ICML Workshop on High-Dimensional Learning Dynamics (HiLD), Journal of Machine Learning Research (JMLR), and Journal of the Royal Statistical Society: Series B (Statistical Methodology).

Technical Skills

- **Programming:** Python, R, MATLAB, SQL, C/C++, C#, Java, CUDA.
- **Tools:** PyTorch, TensorFlow, Git, L^AT_EX, Linux/Ubuntu, Vim, Unity, Origin.

References

- **Prof. Theodor Misiakiewicz**, Department of Statistics and Data Science, Yale University.
theodor.misiakiewicz@yale.edu
- **Prof. Zhou Fan**, Department of Statistics and Data Science, Yale University.
zhou.fan@yale.edu
- **Prof. Weijie Su**, Department of Statistics and Data Science, The Wharton School, University of Pennsylvania.
suw@wharton.upenn.edu
- **Dr. Alberto Bietti**, Center for Computational Mathematics, Flatiron Institute.
abietti@flatironinstitute.org
- **Dr. Denny Wu**, Faculty Fellow, Center for Data Science, New York University; Center for Computational Mathematics, Flatiron Institute.
dennywu@nyu.edu